Technische Universität München
Institute for Astronomical and Physical Geodesy

Adv Orb Mech Tutorial

Exercise 1: Homogeneous Solution of Hill Equations

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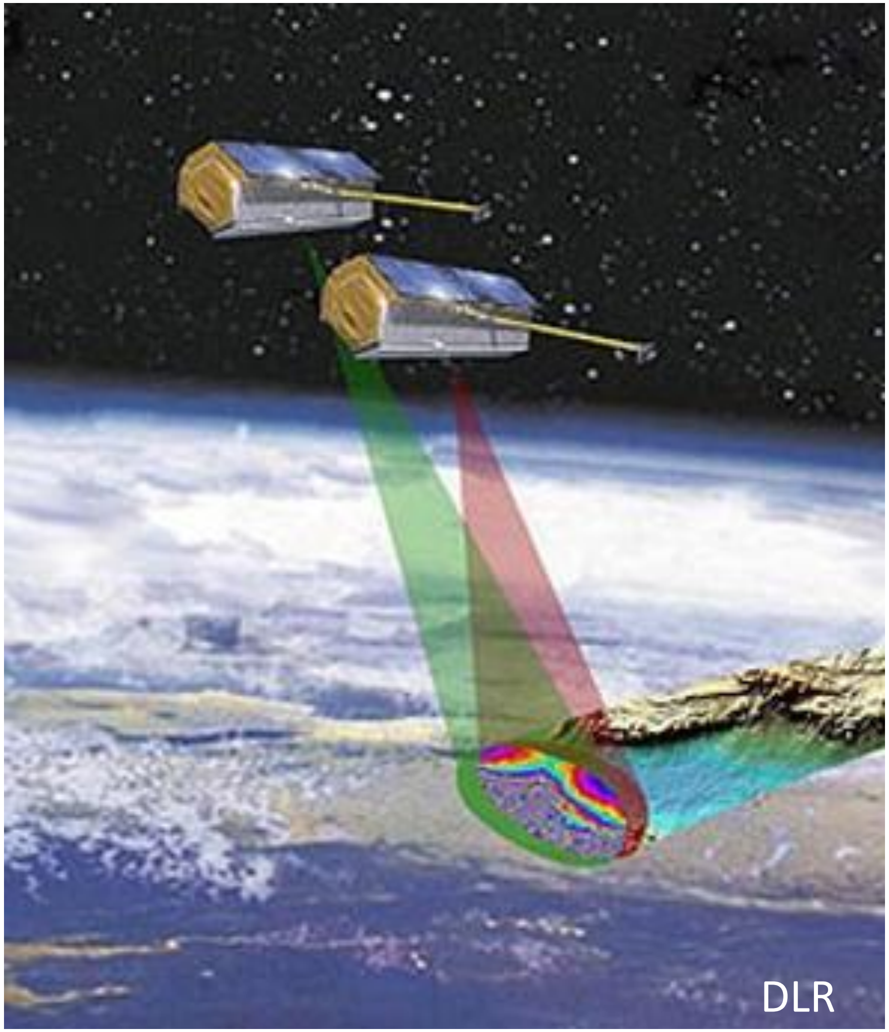
2 tutorials in the semester:

1. Homogeneous Solution of Hill Equation
2. Orbit Perturbations

1. Introduction

TerraSAR and TanDEM-X

- Near circular orbit
- ~500 km height
- Fly very close formation, only several hundred meters
- Deliver high quality elevation data by means of SAR interferometry



2. Task

Task I: Display absolute and relative motion of the two satellites in two frames

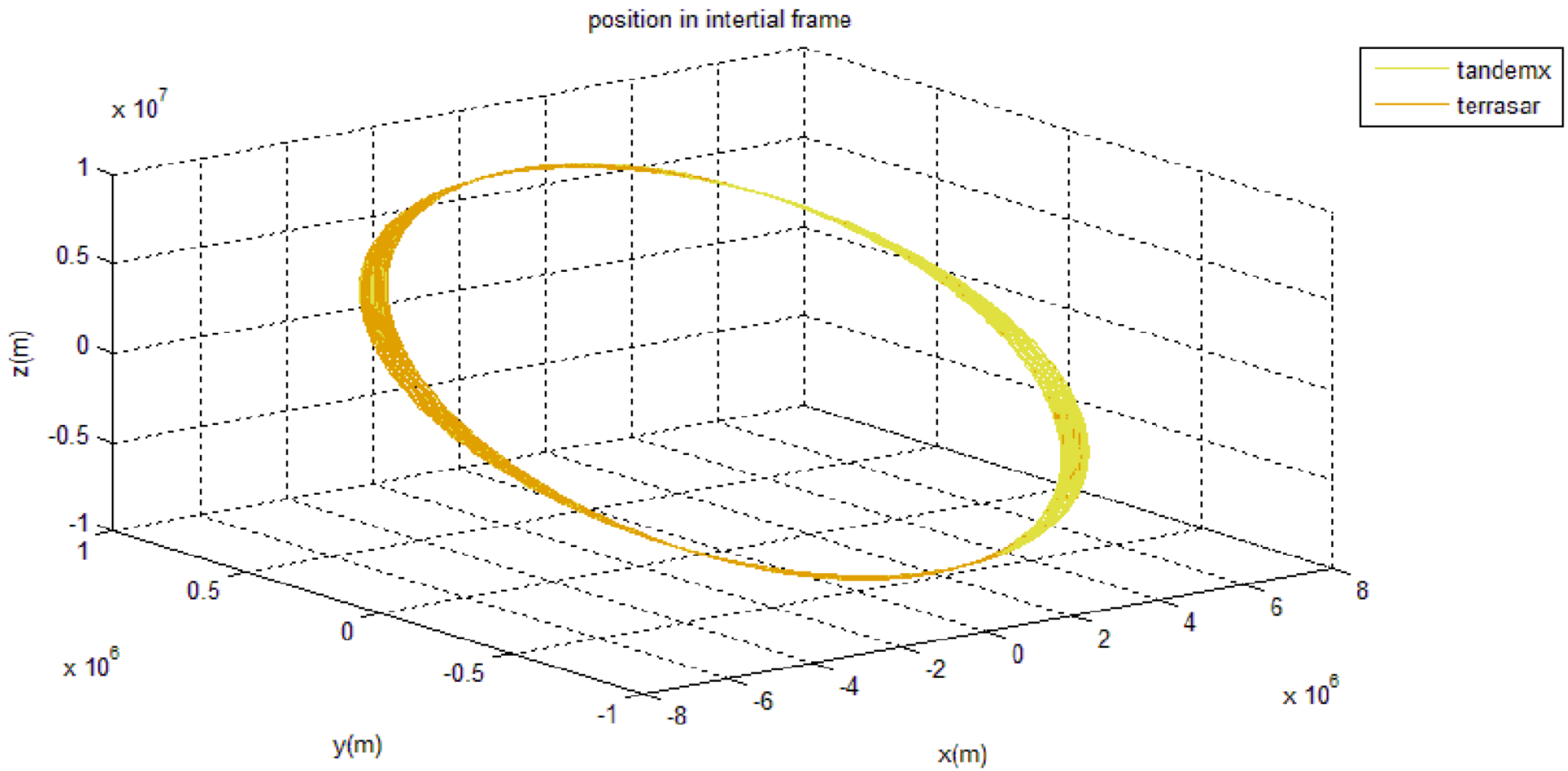
Task II: Assess the difference between the perturbed and the unperturbed motion of TanDEM-X with respect to TerraSAR

Orbit data for TanDem-X and TerraSAR from DLR:

- Positions (columns: 1-3, m) and velocities (columns: 4-6, m/s) with 10 sec sampling in inertial space, covering 30 hours, one file per satellite.

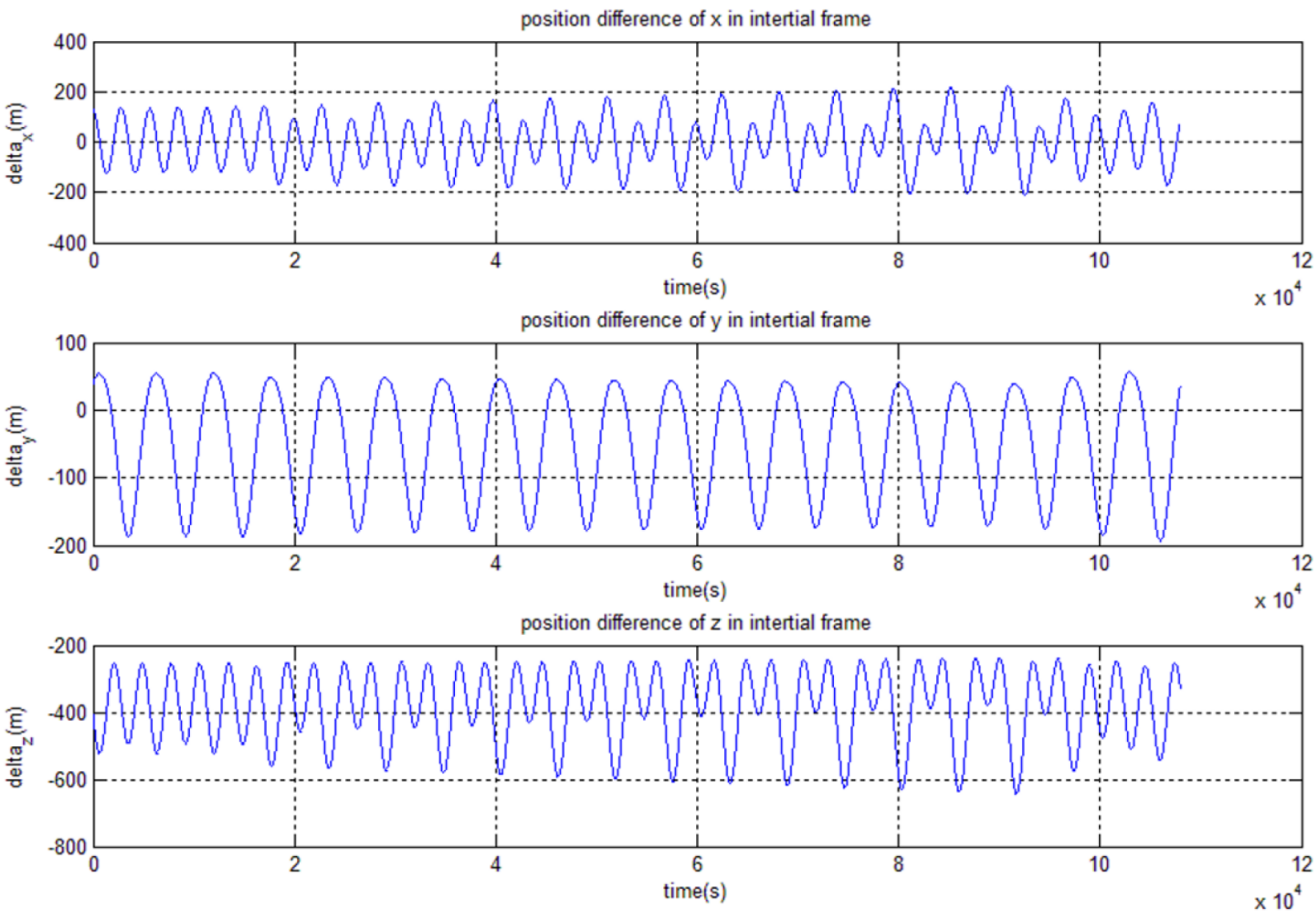
Task I:

1. Display the orbits of the two satellites



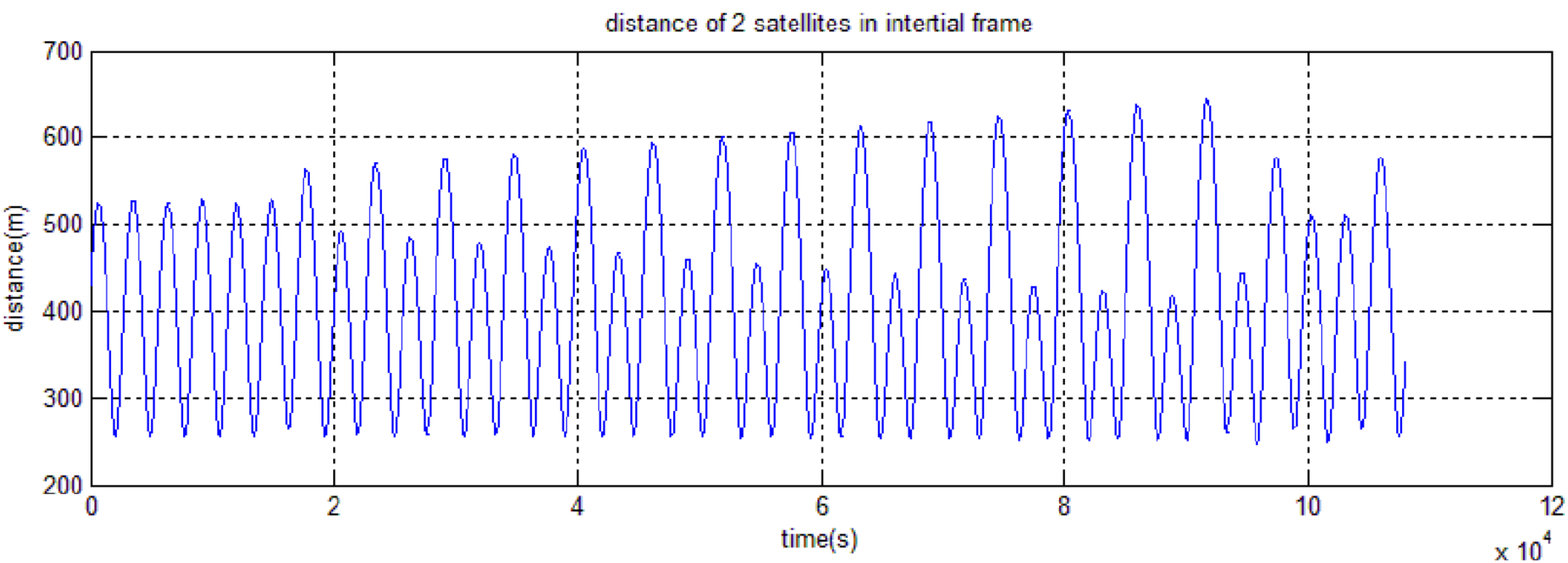
Task I:

Plot the components of the difference vector between the two satellites as a function of time



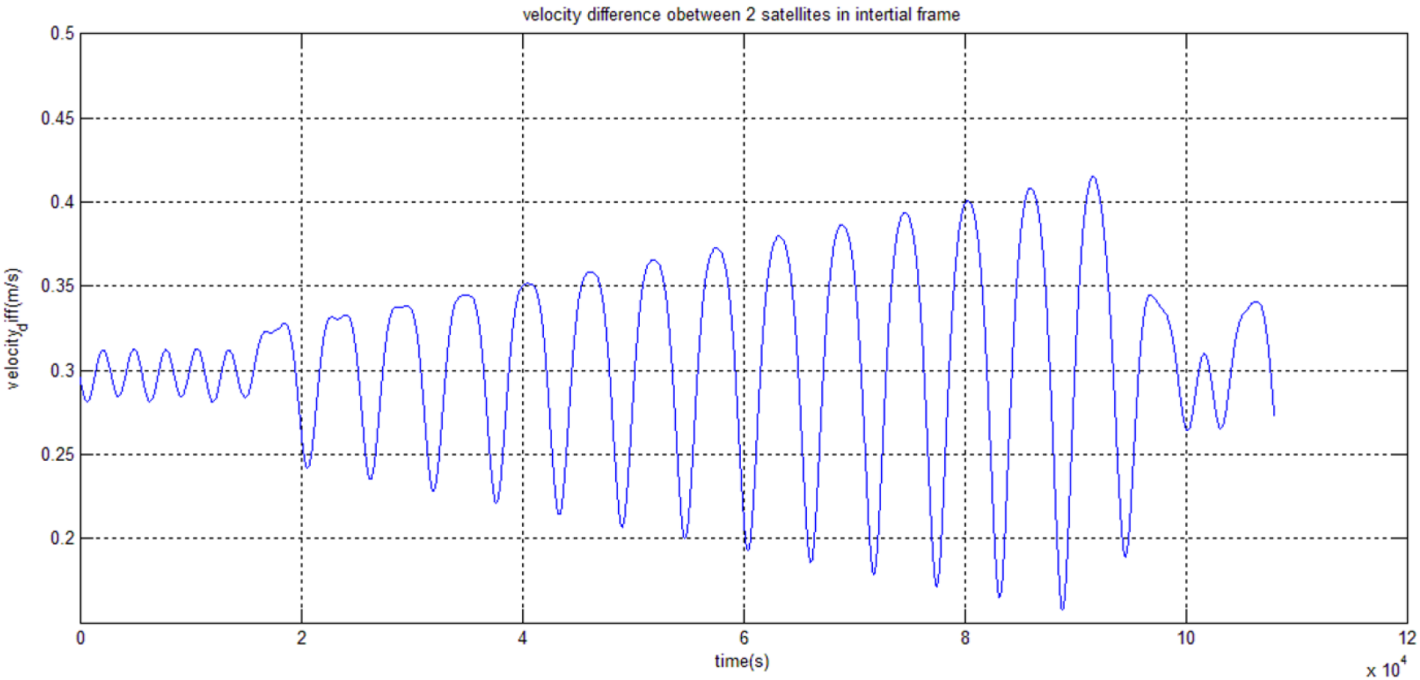
Task I:

Plot the distance between the two satellites as a function of time



Task I:

Plot the absolute value of the velocity difference as a function of time



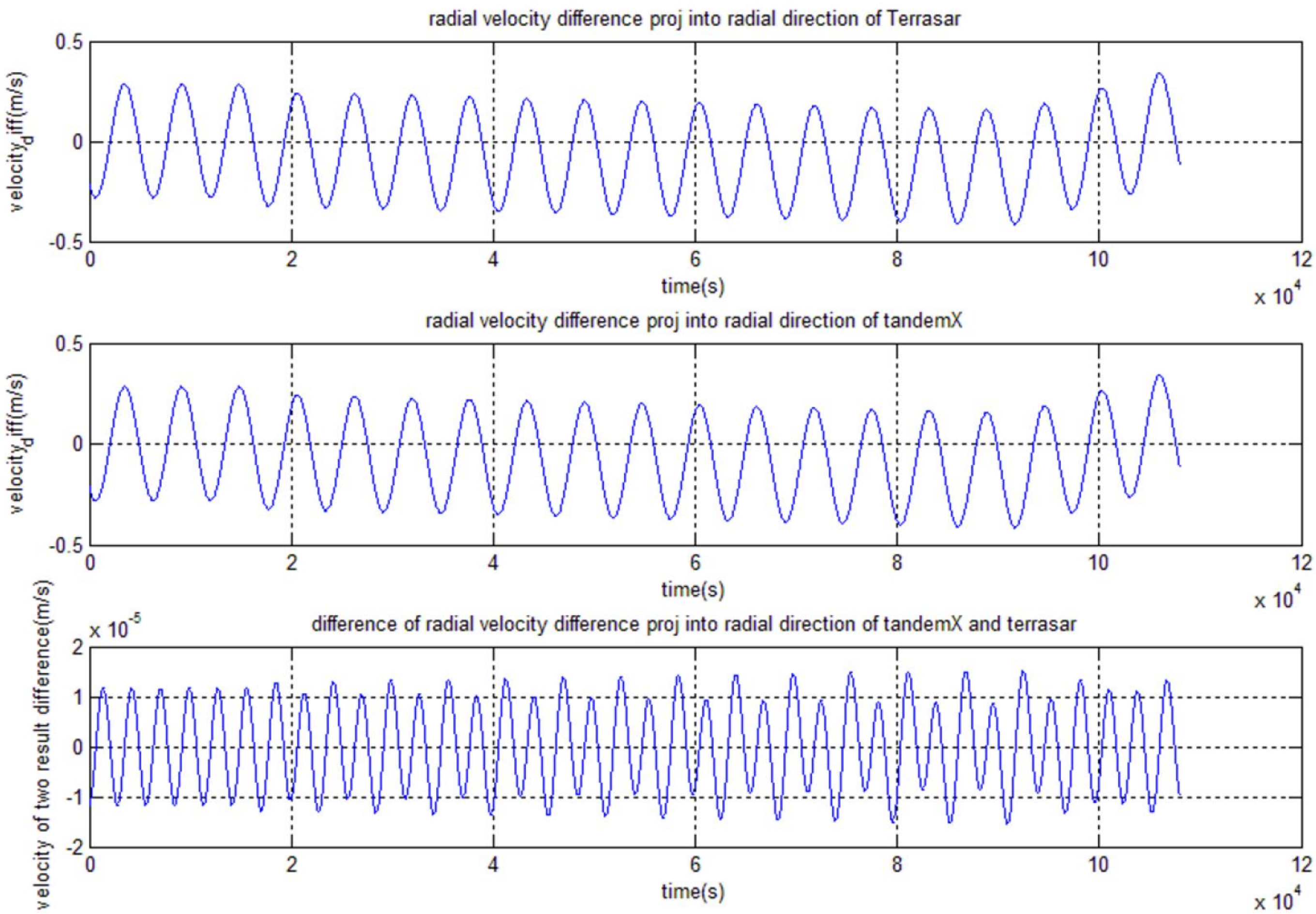
$$v = \sqrt{(v_{x1} - v_{x0})^2 + (v_{y1} - v_{y0})^2 + (v_{z1} - v_{z0})^2}$$

Task I:

Plot the radial velocity as function of time

$$\mathbf{e}_r = \frac{\Delta \mathbf{x}}{|\Delta \mathbf{x}|}$$

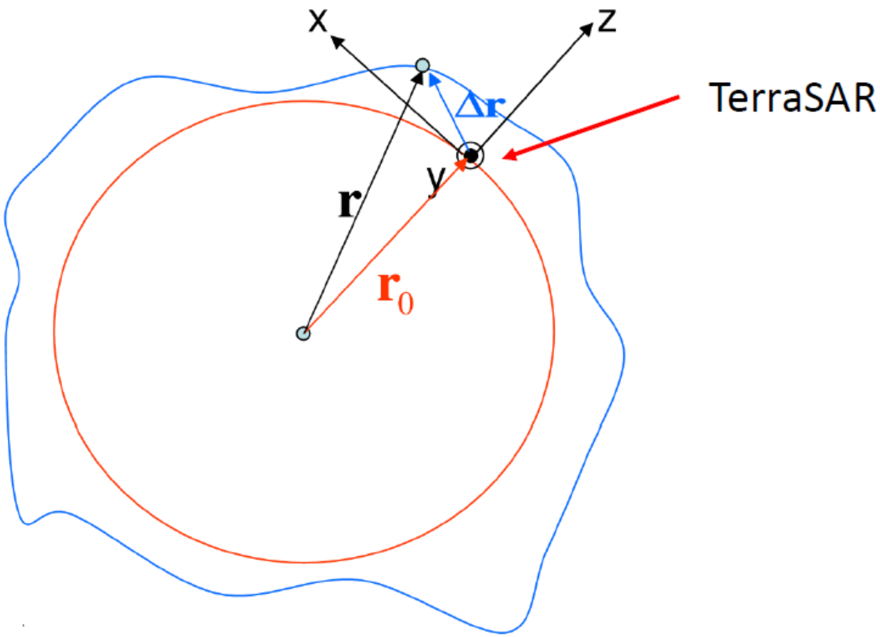
$$v_r = \Delta \mathbf{v} \cdot \mathbf{e}_r$$



Task I:

2. Define the orbit-fixed reference system related to TerraSAR

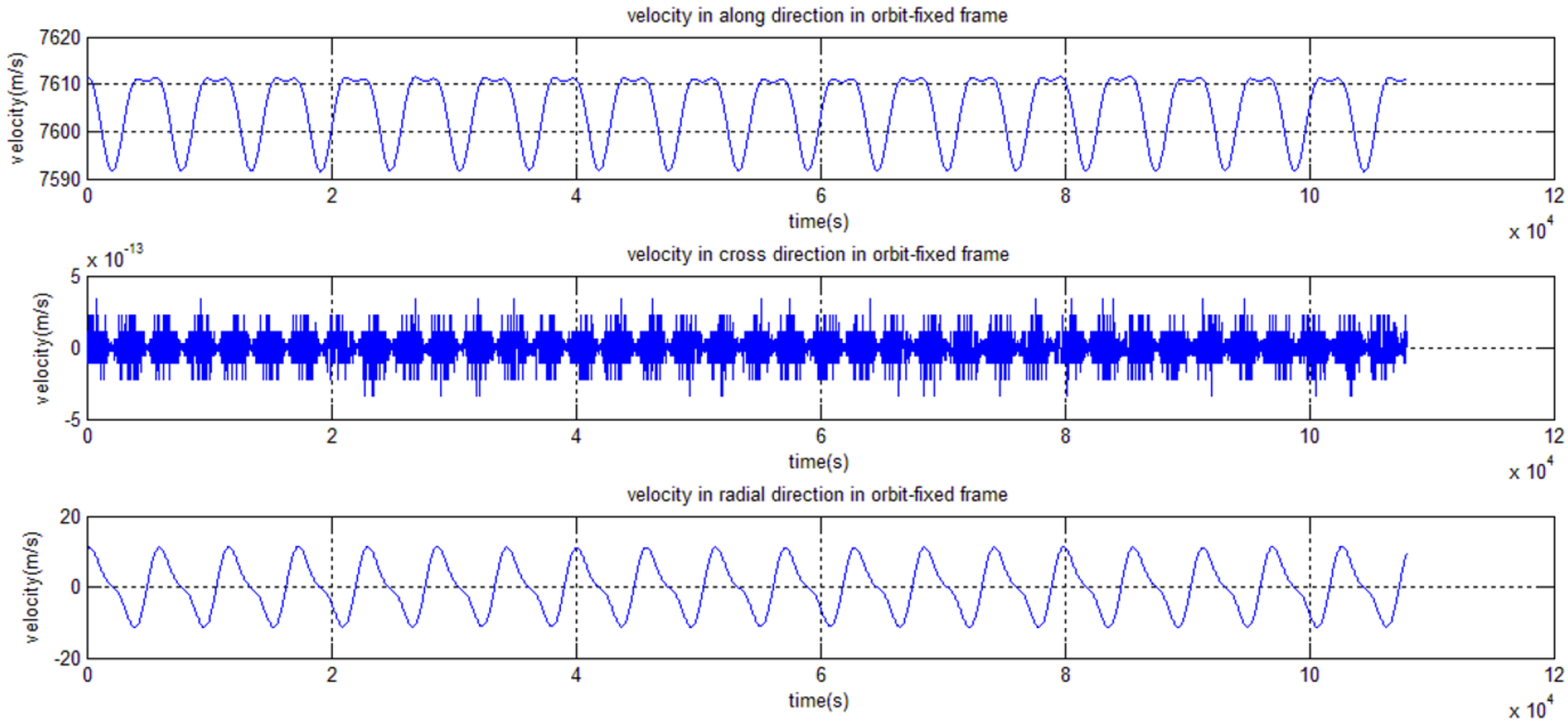
Center: Satellite (TerraSAR)
x: tangential to the circular orbit in the direction of motion (S)
y: perpendicular to plane of the circular orbit (W)
z: radially outward (R)



$$\mathbf{e}_z = \frac{\mathbf{r}}{|\mathbf{r}|} \qquad \mathbf{e}_y = \frac{\mathbf{r}\times\mathbf{v}}{|\mathbf{r}\times\mathbf{v}|} \qquad \mathbf{e}_x = \mathbf{e}_y \times \mathbf{e}_z$$

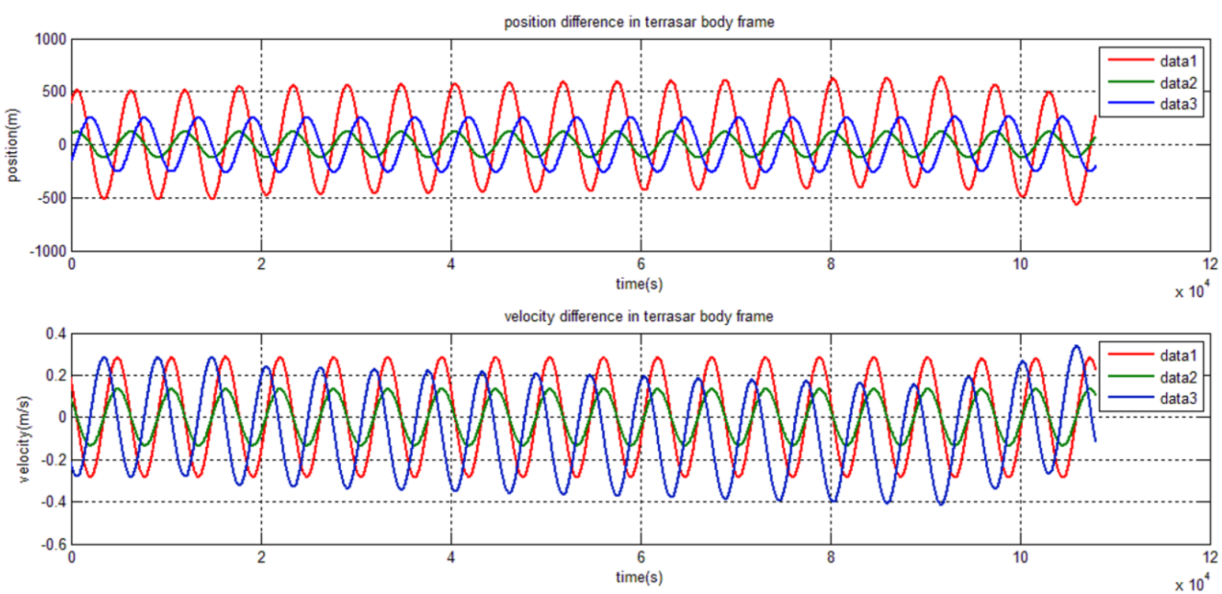
Task I:

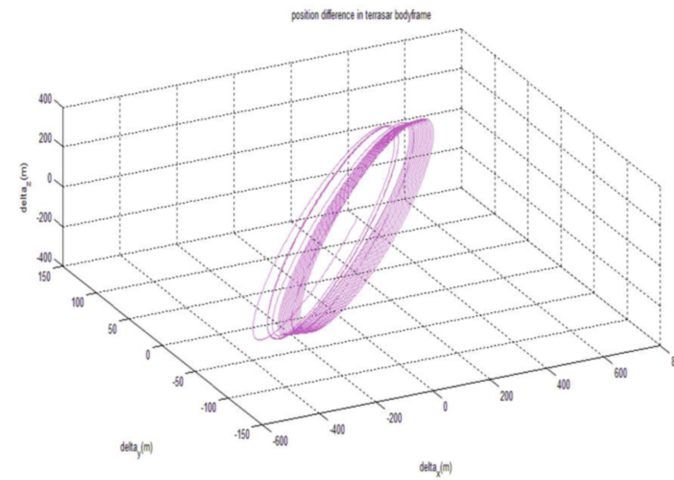
Plot the components of the velocity vector of TerraSAR in this reference frame and discuss the result



Task I:

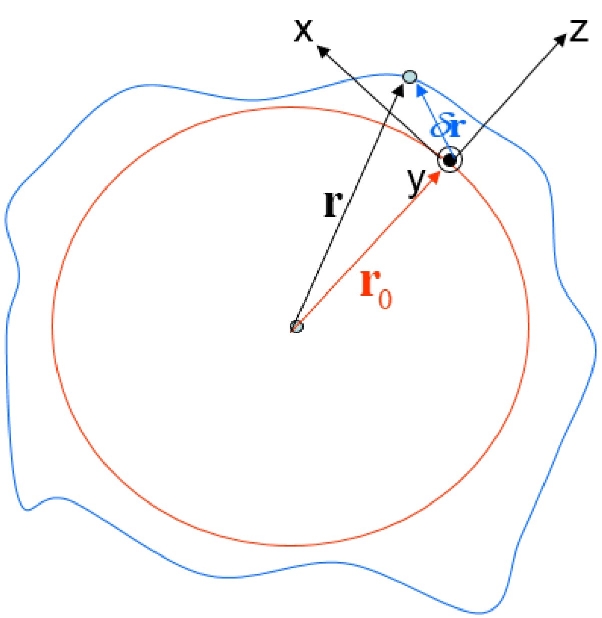
3. Display the position difference and the velocity difference vector of TanDEM-X in the same reference frame. Display the orbit of TanDEM-X around TerraSAR





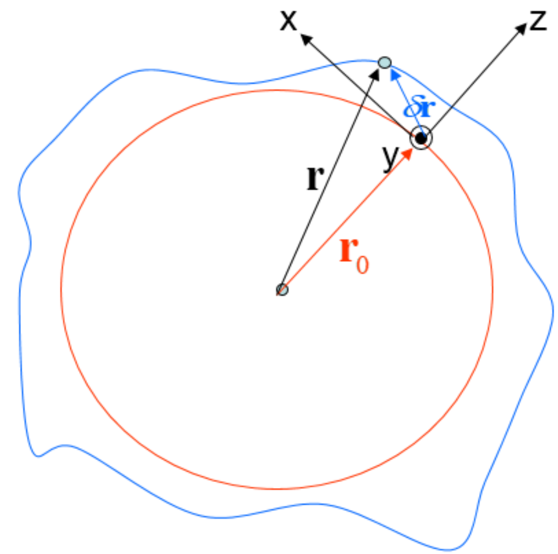
Task II:

- **Assess the difference between the perturbed and the unperturbed motion of TanDEM-X w.r.t. TerraSAR**
- Perturbed motion is calculated in Task I
- Unperturbed motion ins calculated by the Hill equations
- Hill equations describe the relative motion between the true and the approximate circular orbit
- Here the true orbit is unperturbed TanDEM-X, circular orbit is TerraSAR



Task II:

- Solution of Hill equations



$$x(t) = \frac{2}{\bar{n}}\dot{z}_0 \cos \bar{n}t + \left(\frac{4}{\bar{n}}\dot{x}_0 + 6z_0\right)\sin \bar{n}t - \left(3\dot{x}_0 + 6\bar{n}z_0\right)t + x_0 - \frac{2}{\bar{n}}\dot{z}_0$$

$$y(t) = y_0 \cos \bar{n}t + \frac{\dot{y}_0}{\bar{n}}\sin \bar{n}t$$

$$z(t) = \left(-\frac{2}{\bar{n}}\dot{x}_0 - 3z_0\right)\cos \bar{n}t + \frac{\dot{z}_0}{\bar{n}}\sin \bar{n}t + \frac{2}{\bar{n}}\dot{x}_0 + 4z_0$$

$$\dot{x}(t) = -2\dot{z}_0 \sin \bar{n}t + (4\dot{x}_0 + 6\bar{n}z_0)\cos \bar{n}t - 3\dot{x}_0 - 6\bar{n}z_0$$

$$\dot{y}(t) = -y_0\bar{n}\sin \bar{n}t + \dot{y}_0 \cos \bar{n}t$$

$$\dot{z}(t) = (2\dot{x}_0 + 3\bar{n}z_0)\sin \bar{n}t + \dot{z}_0 \cos \bar{n}t$$

known: $x(t), y(t), z(t)$

21003 equations (3 coords for epochs 2000:9000)

unknown: $x_0, y_0, z_0, \dot{x}_0, \dot{y}_0, \dot{z}_0$

6 unknowns

→ problem solved

Task II:

- Step 1: Solve 6 initial values using Least Squares

$$(x_0, \dot{x}_0, z_0, \dot{z}_0)^T = \left(\mathbf{A}_{xz}^T \mathbf{A}_{xz}\right)^{-1} \mathbf{A}_{xz}^T (x, z)^T$$

$$(y_0, \dot{y}_0) = \left(\mathbf{A}_y^T \mathbf{A}_y\right)^{-1} \mathbf{A}_y^T y$$

$$\mathbf{A}_{xz} = \begin{pmatrix} \frac{\partial x}{\partial x_0} & \frac{\partial x}{\partial \dot{x}_0} & \frac{\partial x}{\partial z_0} & \frac{\partial x}{\partial \dot{z}_0} \\ \frac{\partial z}{\partial x_0} & \frac{\partial z}{\partial \dot{x}_0} & \frac{\partial z}{\partial z_0} & \frac{\partial z}{\partial \dot{z}_0} \end{pmatrix} \; and \; \mathbf{A}_y = \begin{pmatrix} \frac{\partial y}{\partial y_0} & \frac{\partial y}{\partial \dot{y}_0} \end{pmatrix}$$

Repeat 2 x 4 matrix for each epoch to obtain 14002 x 4 matrix

14002×4

Repeat 1 x 2 matrix for each epoch to obtain 70021x 2 matrix

7001×2

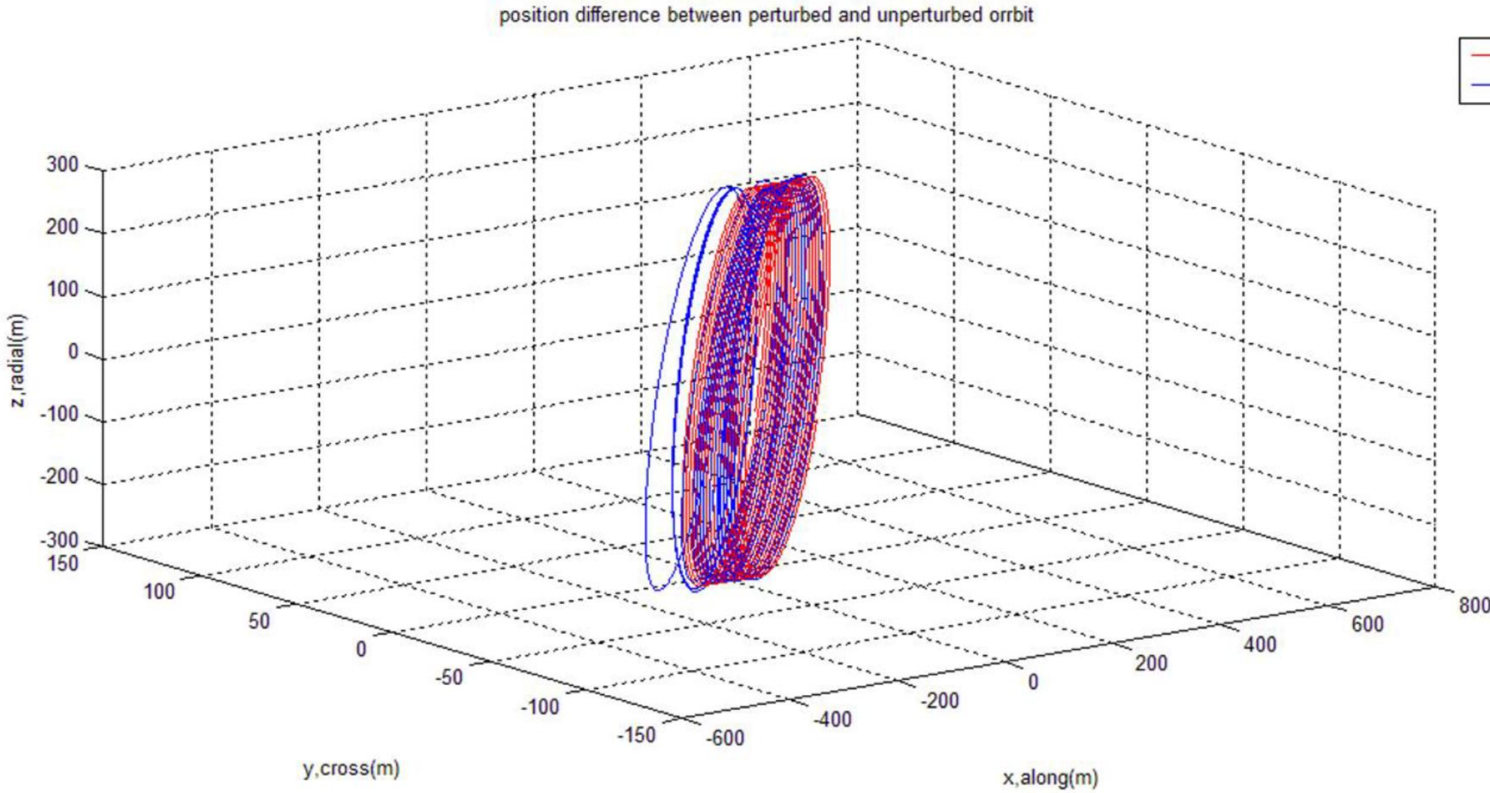
$\mathbf{A}$ is a function of time and mean motion

Show initial values in your report!

- Step 2: Plug 6 initial values into Hill equations and retrieve $x(t), y(t), z(t)$.

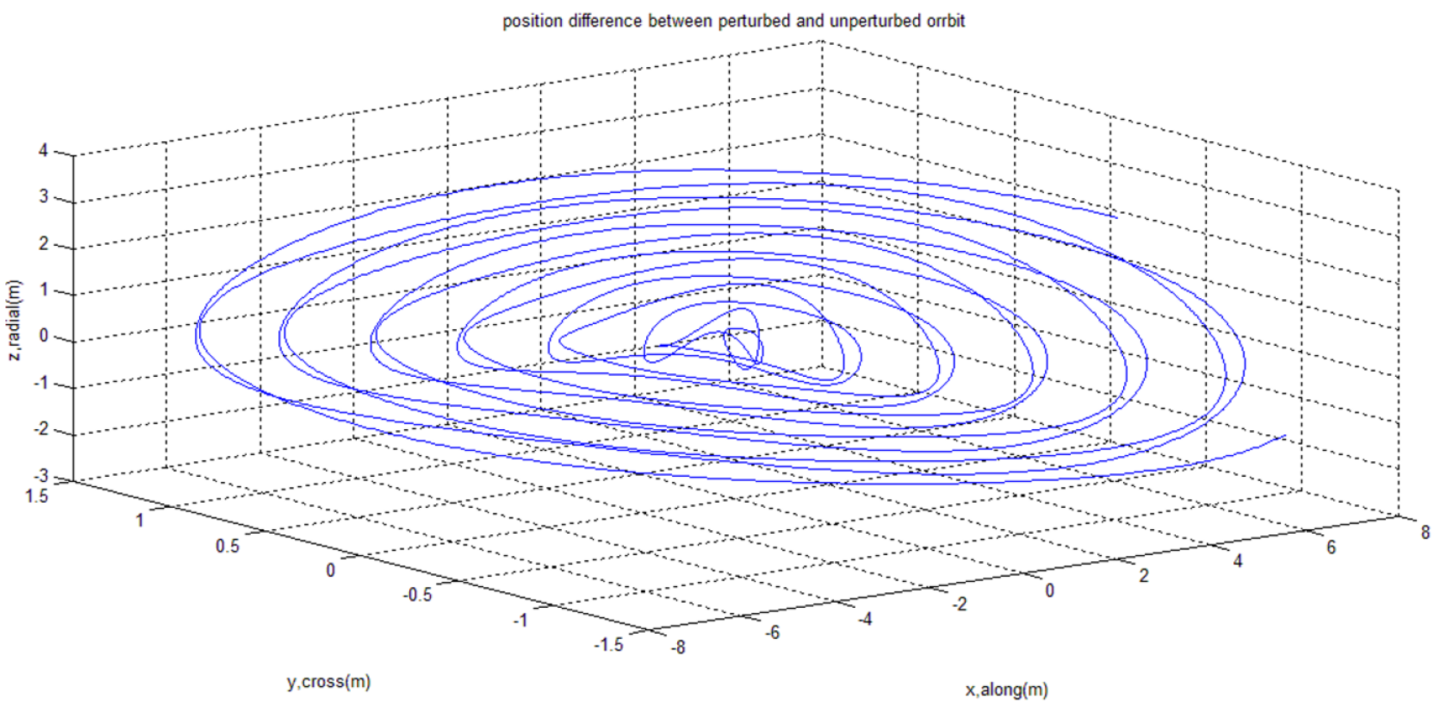
Task II:

Step 3. Show the unperturbed relative motion derived from the Hill equation



Task II:

Compare this solution with perturbed case in Task I



Of course adding position difference in 2D profiles or velocity differences are also welcome

Prepare a short report and discuss the results

Add Matlab code as appendix

Submission of report until June 08, 2026